

MAT 102: Ordinary Differential Equations

Topic 3: Second Order Linear Ordinary Differential Equations

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1 Tutorial Questions — Topic 3

Instructions

Show all working clearly. Identify the method used for each problem. State the general solution before applying any conditions.

1.1 Section A: Linear Independence and the Wronskian

Q1. Compute the Wronskian of each pair of functions and determine whether they are linearly independent:

(a) $y_1 = e^{3x}, y_2 = e^{-x}$

(b) $y_1 = \cos 3x, y_2 = \sin 3x$

(c) $y_1 = x^2, y_2 = x^3$

(d) $y_1 = e^x, y_2 = e^{x+2}$

(e) $y_1 = e^{2x} \cos x, y_2 = e^{2x} \sin x$

(f) $y_1 = 1 + x, y_2 = 1 - x$

Q2. Verify that y_1 and y_2 are solutions of the given ODE, compute their Wronskian, and write the general solution.

(a) $y_1 = e^{-2x}, y_2 = e^{3x}$; ODE: $y'' - y' - 6y = 0$

(b) $y_1 = x, y_2 = xe^x$; ODE: $x^2y'' - (x^2 + 2x)y' + (x + 2)y = 0, x > 0$

(c) $y_1 = e^x \cos 2x, y_2 = e^x \sin 2x$; ODE: $y'' - 2y' + 5y = 0$

1.2 Section B: Homogeneous Equations — Constant Coefficients

Q3. Solve the following homogeneous ODEs:

(a) $y'' - 7y' + 12y = 0$

(b) $y'' + 6y' + 9y = 0$

(c) $y'' - 4y' + 13y = 0$

(d) $y'' - 9y = 0$

(e) $y'' + 2y' + y = 0$

(f) $4y'' - 4y' + y = 0$

(g) $y'' + \omega^2 y = 0$ (simple harmonic oscillator, $\omega > 0$)

(h) $y'' + 2y' - 8y = 0$

Q4. Solve the IVPs:

(a) $y'' + 4y = 0, \quad y(0) = 3, y'(0) = -2$

(b) $y'' - 2y' - 3y = 0, \quad y(0) = 2, y'(0) = -2$

(c) $y'' + 6y' + 9y = 0, \quad y(0) = 2, y'(0) = 1$

(d) $y'' - 4y' + 5y = 0, \quad y(0) = 1, y'(0) = 0$

(e) $y'' - 6y' + 9y = 0, \quad y(0) = 0, y'(0) = 2$

1.3 Section C: Method of Undetermined Coefficients

Q5. Find a particular solution y_p using undetermined coefficients:

(a) $y'' - y' - 2y = 3e^{2x}$

(b) $y'' + 4y = 2 \cos x$

(c) $y'' + 2y' - 3y = x^2$

(d) $y'' + y' = 2x^2 - 1$

(e) $y'' - y = e^x \sin x$

(f) $y'' + y = \sin x$ (*watch for modification rule*)

(g) $y'' - 2y' = 4x + 2$

Q6. Solve the following non-homogeneous ODEs completely (find $y_h + y_p$):

(a) $y'' - 5y' + 6y = 2e^x$

(b) $y'' + 4y' + 4y = 3xe^{-2x}$ (*modification rule applies!*)

(c) $y'' - 4y = 4x^2 - 2$

(d) $y'' + 3y' + 2y = \cos x + \sin x$

(e) $y'' - y = e^x + e^{-x}$ (*modification rule applies twice!*)

1.4 Section D: Variation of Parameters

Q7. Use variation of parameters to find the general solution:

(a) $y'' + y = \tan x$

(b) $y'' + 4y = 4 \csc 2x$

(c) $y'' - 2y' + y = \frac{e^x}{x^2}, \quad x > 0$

(d) $y'' - y = \frac{2}{1 + e^x}$

(e) $y'' + 3y' + 2y = \frac{1}{1 + e^x}$

Q8. Solve the following IVPs using variation of parameters:

(a) $y'' - y = 2e^x, \quad y(0) = 0, y'(0) = 1$

(b) $y'' + 4y = \sec 2x, \quad y(0) = 0, y'(0) = 0$

1.5 Section E: Initial and Boundary Value Problems

Q9. Solve the IVPs completely:

(a) $y'' - 3y' + 2y = 4e^x, \quad y(0) = 1, y'(0) = 1$

(b) $y'' + 4y' - 5y = 3e^{2x}, \quad y(0) = 2, y'(0) = 1$

(c) $y'' + y = x \cos x, \quad y(0) = 0, y'(0) = 1$ (*Hint: use undetermined coefficients — modification needed*)

Q10. For each BVP, determine whether there is no solution, a unique solution, or infinitely many solutions. Find the solution if it exists.

(a) $y'' + y = 0, \quad y(0) = 0, y(\pi) = 0$

(b) $y'' + y = 0, \quad y(0) = 1, y'(\pi) = 0$

(c) $y'' + 4y = 0, \quad y(0) = 0, y(\pi) = 2$

(d) $y'' + 4y = 0, \quad y(0) = 0, y(\pi/2) = 0$

(e) $y'' + 9y = 0, \quad y(0) = 0, y(\pi/3) = 1$

1.6 Section F: Applications

Q11. A spring-mass system has mass $m = 2$ kg and spring constant $k = 18$ N/m.

(a) Find the natural frequency ω_0 and the period of oscillation.

(b) If $x(0) = 1$ m and $x'(0) = 3$ m/s, find $x(t)$.

(c) Write the solution in the form $R \cos(\omega_0 t - \delta)$ and find the amplitude R and phase δ .

Q12. A spring-mass-dashpot system has $m = 1$ kg, $b = 4$ kg/s, $k = 4$ N/m, with $x(0) = 1$ m and $x'(0) = 0$.

(a) Write the ODE.

(b) Solve completely and classify the damping.

(c) Find the time at which the mass first returns to the equilibrium $x = 0$.

Q13. A mass of 0.5 kg is attached to a spring with $k = 2$ N/m. The system has a damping force of $2x'$ N. An external force $F(t) = 3 \cos 3t$ N is applied. Initially $x(0) = 0$, $x'(0) = 0$.

- (a) Write the IVP.
 - (b) Find the complementary function x_h .
 - (c) Find the particular solution x_p (steady-state response).
 - (d) Solve the IVP completely.
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Q14. An RLC series circuit has inductance $L = 1$ H, resistance $R = 2 \Omega$, and capacitance $C = \frac{1}{2}$ F. The EMF is $E(t) = 10 \sin t$ V. Initially $q(0) = 0$, $i(0) = q'(0) = 0$.

- (a) Write the ODE for $q(t)$.
 - (b) Solve for $q(t)$.
 - (c) Find the current $i(t) = q'(t)$.
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Q15 (Challenge). Prove that if y_1 and y_2 are solutions of $y'' + P(x)y' + Q(x)y = g(x)$ (non-homogeneous), then $y_1 - y_2$ is a solution of the corresponding homogeneous equation. What does this tell you about the structure of the general solution?
